

WINDHOEK, Namibia

WMO: 681100

Lat: 22.567S

Lon: 17.100E

Elev: 1725

StdP: 82.25

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.0	5.9	-20.6	0.7	19.1	-17.4	1.0	18.3	8.2	14.4	7.1	15.0	3.4	90	0.337

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	13.5	34.2	14.9	33.2	14.6	32.2	14.3	18.7	25.6	18.2	25.1	17.7	24.6	3.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.1	15.1	20.1	16.3	14.4	19.8	15.9	13.9	19.7	61.4	25.5	59.1	25.1	57.2	24.4	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.1	6.2	5.4	DB	1.4	36.5	2.1	1.9	-0.1	37.8	-1.4	38.9	-2.6	39.9	-4.1	41.3	(4)
(5)			WB	-2.4	19.9	1.3	1.6	-3.3	21.0	-4.1	22.0	-4.8	22.9	-5.8	24.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.7	24.1	23.8	22.3	20.2	17.8	15.4	14.9	17.4	21.2	23.3	23.9	24.5	(6)	
	DBStd	4.24	2.35	2.43	2.56	2.23	2.56	2.15	2.49	2.79	2.98	2.77	2.72	2.46	(7)	
	HDD10.0	5	0	0	0	0	1	1	2	1	0	0	0	0	(8)	
	HDD18.3	308	0	1	9	37	91	107	48	10	3	2	1	1	(9)	
	CDD10.0	3917	437	385	382	306	244	161	154	231	337	413	416	451	(10)	
	CDD18.3	1177	179	152	124	64	21	2	1	19	97	157	168	192	(11)	
	CDH23.3	11533	1639	1346	1044	529	238	41	44	263	1027	1644	1741	1976	(12)	
(13)	CDH26.7	4412	699	557	366	94	22	2	2	24	330	674	752	892	(13)	
(14)	Wind	WSAvg	2.4	2.3	2.2	2.2	2.4	2.4	2.6	2.7	2.7	2.8	2.6	2.3	(14)	
(15) Precipitation	PrecAvg	371	76	96	84	33	6	1	0	1	3	12	27	32	(15)	
	PrecMax	701	303	196	177	115	26	14	0	4	27	58	106	118	(16)	
	PrecMin	136	2	4	30	5	0	0	0	0	0	0	0	0	(17)	
	PrecStd	130	75	57	47	33	8	4	0	1	8	15	29	29	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.1	34.5	33.9	30.2	28.8	26.1	26.1	28.5	33.1	34.2	34.9	35.9	(19)	
		MCWB	15.3	15.3	14.9	15.0	12.9	10.7	10.2	11.0	13.8	13.7	14.6	17.5	(20)	
	2%	DB	33.5	33.0	32.0	29.0	27.0	24.3	24.2	27.2	31.1	32.9	33.2	34.1	(21)	
		MCWB	15.0	15.4	14.7	14.1	11.9	9.8	9.3	10.4	12.3	13.1	14.0	15.3	(22)	
	5%	DB	32.1	31.8	30.3	27.9	25.8	23.1	23.2	26.1	29.9	31.8	32.1	32.6	(23)	
		MCWB	15.0	15.6	14.8	13.9	11.7	9.7	9.1	10.0	11.8	12.8	13.7	14.7	(24)	
	10%	DB	30.6	30.1	28.9	26.5	24.7	22.1	22.0	24.9	28.7	30.4	31.0	31.2	(25)	
		MCWB	15.1	15.5	14.9	13.8	11.3	9.3	8.6	9.5	11.3	12.4	13.4	14.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.1	19.6	19.0	17.8	14.4	12.2	11.5	12.4	15.8	15.9	17.5	21.0	(27)	
		MCDB	25.6	25.6	25.5	24.7	23.8	22.9	24.4	26.0	25.2	27.4	25.7	29.4	(28)	
	2%	WB	18.4	18.7	18.1	16.7	13.5	11.1	10.4	11.5	14.0	15.0	16.5	18.6	(29)	
		MCDB	25.2	24.9	24.7	23.6	23.9	22.2	22.5	25.1	27.6	26.7	25.7	27.5	(30)	
	5%	WB	17.9	18.1	17.5	15.9	12.7	10.5	9.6	10.8	13.0	14.3	15.8	17.4	(31)	
		MCDB	24.9	24.7	23.8	23.8	23.7	21.6	21.6	24.3	26.9	26.4	25.3	25.4	(32)	
	10%	WB	17.3	17.5	16.9	15.2	11.9	9.7	8.9	10.0	12.2	13.7	15.2	16.7	(33)	
		MCDB	24.2	24.2	23.1	23.3	23.1	20.8	20.8	23.3	26.0	26.3	25.2	25.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.6	12.3	12.0	12.6	13.3	13.8	14.1	14.5	14.4	14.0	13.8	13.5	(35)	
		MCDBR	14.5	14.3	13.8	13.5	13.9	14.5	14.8	14.9	14.6	14.5	14.6	14.8	(36)	
	5% WB	MCWBR	4.9	5.2	5.0	5.4	6.0	6.6	6.8	6.7	6.4	6.0	5.8	5.4	(37)	
		MCWBR	11.3	11.0	11.1	11.8	13.0	13.4	13.8	13.4	13.1	12.9	12.8	12.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.336	0.335	0.327	0.299	0.266	0.245	0.253	0.298	0.365	0.374	0.321	0.322	(40)	
		taud	2.508	2.532	2.530	2.574	2.610	2.608	2.556	2.396	2.211	2.239	2.473	2.512	(41)	
	Ebn at Noon		1007	995	977	966	963	968	968	950	922	945	1016	1023	(42)	
		Edh at Noon	115	110	105	93	82	78	85	109	142	146	118	114	(43)	
(44) All-Sky Solar Radiation	RadAvg		7.39	6.91	6.25	5.60	5.09	4.69	4.91	5.69	6.63	7.24	7.79	7.92	(44)	
	RadStd		0.49	0.53	0.35	0.35	0.11	0.10	0.08	0.17	0.21	0.30	0.42	0.43	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page